Exercise 3: SQL CASE Statements

1.
```
SELECT
    product_name,
    price,
CASE
    WHEN price > 1000 THEN 'Expensive'
    WHEN price > Between 100 AND 1000 THEN 'Mid-range'
    WHEN price < 100 THEN 'Budget'
END AS price_category
FROM products;
```

Output

product_name	price	price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

2. TABLE: ORDERS

```
SELECT
    customer_name,
    amount,
CASE
    WHEN amount >= 1000 THEN 'High Value'
    WHEN amount Between 500 AND 999.99 THEN 'Medium
    Value'
    ELSE 'Low Value'
END AS order_value_category
FROM orders;
```

Output

Customer_name	amount	Order_value_category
Alice	150.00	Low value
Bob	560.00	Medium value
Charlie	999.99	Medium value
Diana	45.50	Low value
Ethan	1200.00	High value

3. TABLE: Employees

```
SELECT
    emp_name,
    department,
    Salary,
Case
    WHEN department = 'IT' AND Salary > 80000 THEN 'Senior IT'
    WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'
ELSE AS position_level
FRom employees;
```

Output

Emp_name	department	Salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	58000	Staff

4. TABLE: STUDENT

```
SELECT
    Student_Id,
    Student_name,
    Score,
Case
    WHEN Score >= 90 Then 'A'
    WHEN Score Between 80 and 89 Then 'B'
    WHEN Score Between 70 and 79 Then 'C'
    WHEN Score Between 60 and 69 Then 'D'
    ELSE 'F'
END AS grade
FROM Students;
```

Output

Student-id	Student_name	Score	grade
1	Anna	92	A
2	Ben	76	C
3	Cara	59	F
4	David	83	B
5	Ella	68	D

5. TABLE: deliveries

```
SELECT
    delivery_id,
    delivery_time_minutes,
CASE
    WHEN delivery_time_minutes <= 30 Then 'Fast'
    WHEN delivery_time_minutes Between 31 and 60 Then 'on time'
    ELSE 'Late'
END AS performance
FROM deliveries;
```

Output

delivery_id	delivery_time_minutes	performance
1	45	on Time
2	80	Late
3	30	Fast
4	65	late
5	100	late

~~Table: tickets~~
~~ticket_id~~

~~1~~
~~2~~
~~3~~

6. TABLE: TICKETS

```
SELECT
    ticket_id
    issue_type
    priority
CASE
    WHEN priority = 3 THEN 'High'
    WHEN priority = 2 THEN 'Medium'
    WHEN priority = 1 THEN 'Low'
END AS priority_label
FROM tickets
```

Output

ticket_id	issue_type	priority	priority_level
1	Login issue	1	Low
2	Server down	3	High
3	Slow system	2	Medium
4	Email error	2	Medium
5	Password reset	1	Low

7. TABLE; attendance

```
SELECT
    Student_id
    days_present
    total_days
Round ((days_present * 100.0/total_days), 2 AS attendance_percentage,

case
    WHEN (days_present * 100.0 / total_days) >= 90 Then 'Excellent'
  WHEN (day_present * 100.0 / total_days) Between 75 AND 89 THEN
'GOOD'
ELSE 'needs Improvement'
END AS performance
FROM attendance;
```

Output

Student_id	days_present	total_days	attendance	performance
1	45	50	90.0	Excellent
2	30	50	60.0	needs improvement
3	48	50	96.0	Excellent
4	25	50	50.0	Need improvement
5	50	50	100.0	Excellent

8. TABLE : products_inventory

```
SELECT
    product_id,
    stock_qty,
CASE
    WHEN stock_qty = 0 THEN 'out of stock'
    WHEN stock_qty = 5 THEN 'In stock'
    WHEN stock_qty between 1 AND 4 THEN 'Low stock'

ELSE 'In stock'
END AS stock_status
FROM product_inventory;
```

Output

product_id	stock_qty	stock_status
1	5	In stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

9. TABLE : Classes

```
SELECT
    class_id
    subject,
    enrolled_students
CASE
    WHEN enrolled_students >= 25 THEN 'Large'
    WHEN enrolled_students BETWEEN 10 and 24 THEN 'medium'
ELSE 'SMALL'
```

END AS class_size
FROM classes;

Output

class_id	subject	enrolled_students	class_size
1	math	30	Large
2	English	25	Large
3	Science	15	Medium
4	Art	5	Small
5	History	20	Medium

TABLE : PAYMENT

SELECT
payment_id
amount,
payment_method
case
WHEN payment_method = 'cash' and amount >= 200
THEN 'Eligible for discount
ELSE 'Not Eligible
END AS discount_status
FROM payment

Output

payment_id	amount	payment_method	discount_status
1	50.00	card	Not_Eligible
2	200.00	cash	Eligible for discount
3	150.00	card	Not Eligible
4	75.00	Paypal	Not Eligible
5	300.00	cash	Eligible for discount